

Completion Caching: Reusing LLM Answers at Provider Scale

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1 Introduction

Large language model (LLM) serving has two phases. *Prefill* consumes the prompt in one (or a few) parallel forward passes: every prompt token is looked up in the model’s embedding table and then transformed by every layer on the accelerator. *Decode* then emits tokens autoregressively. Each step streams the model weights, and as the context grows the key–value (KV) cache, from high-bandwidth memory. On contemporary accelerators decode is memory-bandwidth bound and has poor model-FLOP utilization (Leviathan et al., 2023; Pope et al., 2023). Section 2 shows that a short chat prefill is often memory-bound as well. Prefill streams the weights once; decode streams them once per output token. For the short-to-medium prompts and medium-to-long answers that dominate consumer chat, decode occupies the chip, sets time-to-last-token, and, once a fleet is provisioned for peak tokens-per-second, sets the bill.

This cost is paid on every request, including requests that are not unique. Search engines observed decades ago that query streams are heavy-tailed: a modest number of intents account for a large fraction of volume, each appearing as many surface forms. Public chat corpora have the same shape. Topic mass concentrates (Zhao et al., 2024; Zheng et al., 2024), and near-paraphrase clusters are common. For many of those clusters the acceptable answers form a basin rather than a point. The queries “How do I reverse a list in Python?”, “python reverse list in place”, and “what’s the idiomatic way to flip a list?” do not require three independently decoded completions.

We ask whether a provider that already sees this traffic (Google, or any other lab operating a frontier chatbot at scale) can avoid paying decode for the head of that distribution. We call the policy *completion caching*. The cached object is a pair $(Q, C(Q))$: a previously seen query and the completion the model produced for it, not a KV block or a prompt prefix. Given a new query Q , the system decides whether some cached completion C^* is an acceptable answer to Q . If so, it returns C^* without decoding. A first decision uses a small retrieval embedder $e(Q)$, a nearest-neighbor lookup, and a gate, cheap enough to run on the host. If that gate is unsure, a serving-model prefill of Q supplies a hidden state $h(Q)$ in the model’s own geometry, and a second gate can still skip decode. If that gate rejects, the model decodes as usual and the new pair may be inserted.

1.1 What is already cached, and what is not

Serving stacks already cache aggressively. The standard caches do not make this decision.

Prefix and paged KV caches (Kwon et al., 2023) reuse attention state when two requests share a token prefix. Prompt-cache products do the same at the API boundary. They reduce prefill, improve time-to-first-token, and make multi-turn and templated workloads cheaper. They do not skip decode. After the shared prefix is restored, the model still writes a new answer, one bandwidth-bound step at a time. Prefill–decode disaggregation (Zhong et al., 2024) places the two phases on different machines, but still pays for decode.

Exact-match response caches (hash the prompt, return the bytes) skip both phases, but only when the user types the same string again, and they are unsafe even then: two users can type “best Italian restaurant near me” and must not receive each other’s answer. Paraphrase without a shared context is the typical portable case; paraphrase with a bound context is a different object, treated in Section 3.2.

Semantic caches such as GPTCache (Bang, 2023), and more recent context-aware variants (Yan et al., 2025; Yu et al., 2025), are the closest prior work. They embed the query, retrieve a stored response by vector similarity, and, on a hit, bypass the model entirely. Production deployments report large latency reductions when the retrieved response is correct. The limitation is the decision rule. Cosine similarity between query embeddings is a proxy for topical relatedness, not for whether a stored completion is an acceptable answer to this question. Near-neighbors in embedding space include near-misses that matter: different versions (“install Python 3.11” versus “3.12”), different entities with similar form (“capital of France” versus “capital of Spain”), and questions that share a topic but not an answer. Application-layer caches also typically treat the LLM as a black box. They do not use the model’s own prefill state. They rarely fall back to a cheaper decode when the match is only partial, and they are seldom accompanied by a hardware cost model that says when a lookup is cheaper than generating.

A third line of work reuses text without claiming the full answer is reusable. Retrieval-based speculative decoding (He et al., 2024) and prompt-lookup decoding draft tokens from a datastore or from the prompt itself; the model still verifies every token. These methods convert redundancy into a speedup with no quality change. They are complementary: a cached completion that is not safe to serve wholesale can still be a strong draft.

Completion caching is closest to a semantic cache in that it tries to skip decode. Unlike those systems, it may spend a prefill (a signal the model already computes) before deciding, and like speculative retrieval it can fall back to using C^* as a draft rather than as an answer. The decision criterion is answer acceptability, not query similarity or prefix equality.

1.2 The idea

Write $C(\cdot)$ for the serving model’s completion map, and write $\mathcal{A}(Q, C)$ for the (unobserved) predicate that C is an acceptable reply to Q . A completion cache is a set of pairs $\{(Q_i, C_i)\}$ with $C_i = C(Q_i)$ at insertion time. On a new query Q the system must either return some C_i for which $\mathcal{A}(Q, C_i)$ holds, or report a miss and decode $C(Q)$.

$\mathcal{A}$ is not computable from Q alone. Treating $\mathcal{A}(Q, C_i)$ as “ Q is close to Q_i in embedding space” is the error mode above. Prefill and retrieval give cheaper signals than embedding proximity alone:

- a retrieval embedding $e(Q)$ from a small dual encoder (tens to hundreds of millions of parameters), enough to retrieve a shortlist and, when the match is high-confidence, to return a cached completion without touching the serving model;
- the serving model’s last-token prefill state $h(Q)$, already computed if decode is about to start, and a query representation in the model’s own geometry;
- lightweight gates on $(e(Q), e(Q_i), h(Q), h(Q_i), C_i)$ that can be trained or calibrated to be conservative: a false hit serves the wrong answer, and is much more expensive than a false miss, which only costs a decode that would have been run anyway.

Those signals suggest a three-tier policy, specified in Section 3. A high-confidence portable hit (no tools, no per-user agent memory) never touches an accelerator. An ambiguous shortlist is resolved by prefilling Q and gating on $h(Q)$; a pass still skips decode. A reject falls through to

ordinary generation, optionally using the retrieved completion as a speculative draft. If tools gather user-specific or otherwise non-universal context (location, weather, account, live search), the filled C_i is not served; the request may still skip decode by replaying a cached tool plan into a completion template.

Two mechanisms change the hit-rate calculus. *Top-k projective pooling* treats the cache as a small set of completions that can be selected among, voted on, or (in a more speculative design) fused. Many intents have several acceptable answers; requiring a unique nearest neighbor leaves hits on the table. *Gating* keeps the served set inside $\mathcal{A}$. The objective is to maximize decode-skips subject to a bound on unacceptable serves.

1.3 When it pays at provider scale

Completion caching is useful only if decode is expensive relative to lookup and if enough traffic is repeated under a conservative gate. For a large consumer chatbot both hold.

Decode is the scarce resource. Section 2 makes this quantitative across GPU and TPU parts. Prefill of a hundred-token question is one parallel pass. Decode of a three-hundred-token answer is hundreds of sequential, bandwidth-bound steps. Batching amortizes weight traffic across requests and is the main reason serving is viable; it does not make skipped decode free, because a skipped request is a slot that another request can occupy. At fleet scale, hit rate times decode cost is accelerator-hours.

The lookup is cheap relative to that cost. Embedding a short query and searching an HNSW or IVF index of 10^7 – 10^9 vectors is a CPU-and-DRAM problem, typically milliseconds, and is straightforward to colocate with the request router. Returning a few hundred tokens of cached text is a network problem measured in the same units. The relevant comparison is whether retrieval plus a conservative gate is cheaper than occupying an HBM-bound chip for the length of the answer. For typical chat lengths the inequality holds, provided the hit is a true accept.

The traffic is skewed enough that a conservative gate still sees volume. We do not need, and should not claim, that most queries are duplicates. We need the head of the intent distribution (how-to questions, definitions, boilerplate code, exam-style problems, recipes, consumer product questions) to be large in absolute terms. A provider serving hundreds of thousands of queries per second can justify a substantial cache engineering effort at a single-digit-to-low-double-digit hit rate, if those hits are concentrated on decode-heavy, low-personalization requests. The long tail still decodes, as in any cache that only stores the head of a heavy-tailed workload.

The same conditions mark where the idea does not pay. Long-context workloads whose cost is prefill (summarize this document; answer from this unique corpus) gain little from a completion cache: the query is unique and decode may be the smaller term. Deeply personalized or tool-using turns (“what’s on my calendar”; “refactor *this* file”) are not acceptable to serve from someone else’s C_i . Time-sensitive questions go stale. A model-version bump invalidates the cache, or at least requires a freshness policy. These constrain the eligible population; they do not remove the head of the distribution.

1.4 What this paper does

This is a design paper. We do not yet report a production deployment. The contribution is to treat completion reuse as a first-class serving decision and to work out whether, and under what policy, it pays.

Section 2 develops a napkin cost model for prefill and decode on GPU and TPU, including the host and network terms a cache introduces. Section 3 specifies the method: the serve path, gating,

Table 1: Work on a short chat question. The retrieval embedder $e(Q)$ is a separate small model. Prefill and decode use the serving Transformer.

Computation	Work	Device	Order of FLOPs ($S=128$)
Tokenizer	string $\rightarrow$ ids	host CPU	none (table lookup)
Retrieval embedder $e(Q)$	small Transformer, $P_e \sim 2 \times 10^7$	host CPU	$\sim 6 \times 10^9$
Serving prefill $h(Q)$	Llama 3.1 70B, all L layers	GPU/TPU	$\sim 1.8 \times 10^{13}$
One decode step	same 70B weights, one new token	GPU/TPU	$\sim 1.4 \times 10^{11}$

top- k projective pooling, and the miss-side draft and insert rules. Section ?? translates hit rate and gate conservatism into fleet-level savings.

At provider scale, a non-trivial fraction of chat decode is spent writing answers that already exist in the fleet’s own history. A conservative decision, made mostly on CPU, with prefill as a refinement rather than a commitment to generate, can skip that decode.

2 Prefill, decode, and a cost model

Completion caching decides whether to return a stored answer before, during, or after the serving model’s forward pass. The cost of each option depends on what that pass does. This section accounts for the work on a public dense stand-in, Llama 3.1 70B (Grattafiori et al., 2024), whose width, depth, and grouped-query layout are published. A frontier Gemini-class model is wider and deeper; the same identities apply, with different P , L , and d .

2.1 Computations on the path to an answer

A request Q can incur three computations before any completion token is written.

Tokenizer. A byte-pair or SentencePiece tokenizer maps the UTF-8 string Q to integer ids $t_1, \dots, t_S \in \{1, \dots, V\}$. This is a host-side finite-state walk: no Transformer weights, well under a millisecond on a short question, and irrelevant to the cost model.

Retrieval embedder $e(Q)$. A small dual encoder (MiniLM-class, on the order of 2×10^7 parameters, or a larger e5/GTE-class model of a few times 10^8) produces a single vector for nearest-neighbor lookup in the completion cache. It is a Transformer, but not the serving model. Its forward pass is two to three orders of magnitude cheaper than a 70B prefill (Table 1), so it can live on the host next to the request router. A provider that prefers to run it on a small accelerator does not change that comparison.

Serving-model prefill. The serving model reads the same token ids, gathers rows of its embedding table $E \in \mathbb{R}^{V \times d}$, and pushes the resulting $S \times d$ residual stream through all L decoder blocks (Pope et al., 2023). Every linear layer participates. The pass writes a KV cache into HBM and, if generation is about to start, may apply the language-model head $W_{\text{lm}} \in \mathbb{R}^{d \times V}$ to the last position to obtain first-token logits. A cache gate that reads the last-token hidden state $h(Q) \in \mathbb{R}^d$ does not need W_{lm} .

The last two rows determine the relative cost. Prefill of a 128-token question does about as much arithmetic as 126 decode steps, because it applies the body weights to 128 positions at once.

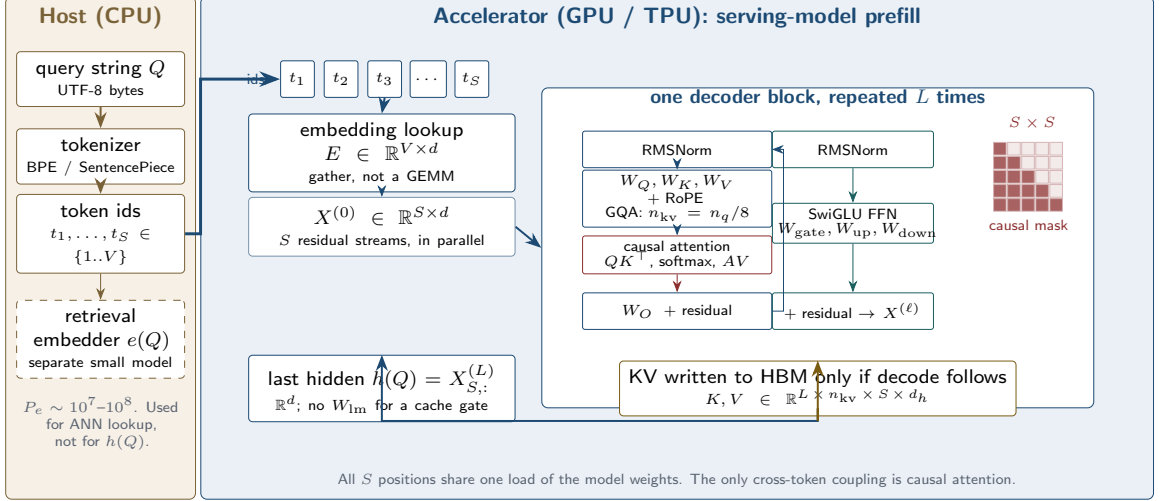


Figure 1: Anatomy of serving-model prefill. The host only tokenizes (and, optionally, runs a small retrieval embedder $e(Q)$). On the accelerator, token ids become an $S \times d$ residual stream by embedding lookup. Each of the L decoder blocks applies RMSNorm, grouped-query projections, rotary positions, a causal $S \times S$ attention, an output projection, and a SwiGLU feed-forward, with residuals around both sub-layers. All S positions are transformed in one weight load. The cache gate reads the last row $h(Q)$; the language-model head is unnecessary for that decision. The KV tensors are written to HBM only if decode will follow.

It does not take 126 times as long. Both phases are dominated by streaming the same ~ 140 GB of BF16 weights off HBM; prefill does it once, decode does it once per emitted token.

2.2 What the serving model does in prefill

Figure 1 is that pass for a Llama-3-style decoder (Ainslie et al., 2023; Grattafiori et al., 2024).

Let d be the residual width, n_q the number of query heads, n_{kv} the number of key/value heads, $d_h = d/n_q$ the head dimension, and d_{ff} the SwiGLU inner width. For Llama 3.1 70B these are

$$L = 80, \quad d = 8192, \quad n_q = 64, \quad n_{kv} = 8, \quad d_h = 128, \quad d_{ff} = 28672, \quad V = 128256.$$

One block, at every position $s = 1, \dots, S$ and in parallel across s , computes

$$\begin{aligned} \tilde{x}_s &= \text{RMSNorm}(x_s), \\ Q_s &= \tilde{x}_s W_Q, \quad K_s = \tilde{x}_s W_K, \quad V_s = \tilde{x}_s W_V, \\ Q_s, K_s &\leftarrow \text{RoPE}(Q_s, K_s), \\ a_s &= \text{softmax}\left(\frac{Q_s K_{\leq s}^\top}{\sqrt{d_h}}\right) V_{\leq s}, \\ x_s &\leftarrow x_s + a_s W_O, \\ x_s &\leftarrow x_s + \text{SwiGLU}(\text{RMSNorm}(x_s); W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}). \end{aligned} \tag{1}$$

Grouped-query attention (Ainslie et al., 2023) is the only structural peculiarity: W_K and W_V map to $n_{kv}d_h = 1024$ dimensions rather than d , and each KV head is shared by $n_q/n_{kv} = 8$ query heads. That shrinks the KV cache by $8\times$ relative to multi-head attention.

Table 2: Worked FLOP counts for Llama 3.1 70B. Attention is negligible on short chat prompts and only a few percent even at $S = 4096$. Decode of T tokens costs about T times a single-token body pass, plus a cheap walk over the growing KV cache.

Quantity	$S=128$	$S=512$	$S=4096$
Linear FLOPs F_{lin}	1.75×10^{13}	7.01×10^{13}	5.61×10^{14}
Attention FLOPs F_{attn}	4.29×10^{10}	6.87×10^{11}	4.40×10^{13}
Prefill total (no lm-head)	1.76×10^{13}	7.08×10^{13}	6.05×10^{14}
Attention share of prefill	0.24%	0.97%	7.3%
One decode step at $c=S$	1.39×10^{11}	1.40×10^{11}	1.50×10^{11}
$T=256$ decode steps, c from S to $S+T$	3.57×10^{13}	3.59×10^{13}	3.83×10^{13}

Two implementation facts matter for caching. First, the serving model’s embedding table E is a gather on the accelerator, $t_s \mapsto E_{t_s}$, and is then dominated by the L blocks that follow; it is not a substitute for the retrieval embedder $e(Q)$. Second, because every position is materialized, prefill emits $h(Q) = X_{S,:}^{(L)}$ with no extra pass. If the gate accepts a cached completion, the newly written KV cache is discarded and decode never starts. If the gate rejects, that KV cache is exactly what decode would have built, so the prefill is not wasted.

2.3 Concrete arithmetic on Llama 3.1 70B

A matrix multiply $Y = XW$ with $X \in \mathbb{R}^{S \times d_{\text{in}}}$ and $W \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ costs $2Sd_{\text{in}}d_{\text{out}}$ FLOPs (multiply-add). Summing the seven linear maps in (1) over L layers gives the body parameter count and the linear FLOPs:

$$\begin{aligned}
P_{\text{attn}} &= d(n_q d_h + 2n_{\text{kv}} d_h + d) = 1.510 \times 10^8, \\
P_{\text{ffn}} &= 3d d_{\text{ff}} = 7.046 \times 10^8, \\
P_{\text{body}} &= L(P_{\text{attn}} + P_{\text{ffn}}) = 6.845 \times 10^{10}, \\
F_{\text{lin}}(S) &= 2P_{\text{body}} S.
\end{aligned} \tag{2}$$

The embedding table and the untied lm-head add $2Vd = 2.101 \times 10^9$ parameters, bringing the public 70.6B total (Grattafiori et al., 2024). Causal attention adds a quadratic term that is still small at chat lengths:

$$F_{\text{attn}}(S) = 4L n_q d_h S^2 = 2.621 \times 10^6 S^2. \tag{3}$$

RMSNorm, SiLU, softmax, and RoPE are $O(LSd)$ and do not move any table below. For a cache gate we omit W_{lm} , so

$$F_{\text{prefill}}(S) = F_{\text{lin}}(S) + F_{\text{attn}}(S) = 1.369 \times 10^{11} S + 2.621 \times 10^6 S^2. \tag{4}$$

One decode step at context length c , *including* the lm-head on the new token, is

$$F_{\text{dec}}(c) = 2P_{\text{body}} + 4L n_q d_h c + 2dV \approx 1.390 \times 10^{11} + 2.621 \times 10^6 c. \tag{5}$$

Table 2 is the FLOP picture. A 128-token prefill is 17.6 TFLOP. A 256-token completion is about 36 TFLOP. Counting FLOPs alone, decode is only twice prefill, which would suggest that skipping decode is a $2\times$ win. That comparison is misleading: the two phases do not have the same arithmetic intensity.

Table 3: Peak arithmetic, HBM bandwidth, and ridge point. H100 BF16 is the *dense* tensor-core rate; NVIDIA’s advertised 1979 TFLOP/s includes 2:4 sparsity, which ordinary dense checkpoints do not use. TPU numbers are per chip (Austin et al., 2025).

Part	Peak (BF16)	Peak (FP8/INT8)	HBM BW	I^* (BF16)
NVIDIA H100 SXM	989 TFLOP/s	1979 TFLOP/s	3.35 TB/s	295 FLOP/B
TPU v5e	197 TFLOP/s	394 TFLOP/s	0.82 TB/s	240 FLOP/B
TPU v5p	459 TFLOP/s	918 TFLOP/s	2.8 TB/s	164 FLOP/B
TPU v6e	920 TFLOP/s	1840 TFLOP/s	1.6 TB/s	575 FLOP/B

Weight traffic, not FLOP count. In BF16 the body plus embed plus head occupy

$$B_w = 2P \approx 1.41 \times 10^{11} \text{ bytes}$$

(141 GB). Prefill reads those weights once. Decode reads them once *per output token*. The KV cache is a rounding error at chat length: K and V together occupy $4Ln_{kv}d_hS = 3.28 \times 10^5 S$ bytes, or 40 MB at $S = 128$ and 1.3 GB at $S = 4096$. Activation traffic for the residual stream is $2Sd$ bytes per layer crossing, also small against 141 GB.

The roofline intensity against weight traffic, for batch B and a multiply-add of $2P$ FLOPs per token, is therefore just the number of tokens that share a weight load (Pope et al., 2023; Williams et al., 2009):

$$I_{\text{prefill}} = BS, \quad I_{\text{dec}} = B \quad (\text{FLOP/byte, BF16}). \quad (6)$$

A chip is compute-bound when I exceeds its ridge point $I^* = \text{peak FLOP/s/HBM bytes/s}$, and memory-bound otherwise.

On an H100, $I^* \approx 295$ in dense BF16 (Table 3). Equation (6) then says:

- Decode is memory-bound until the decode batch approaches $B \approx 300$. A 70B KV cache will not let a single chip get there.
- Prefill of a short question is also memory-bound at batch one: $S = 128 < 295$. It becomes compute-bound once $BS \gtrsim 295$, for example $B = 3$ at $S = 128$, or $B = 1$ at $S = 512$.

The usual claim that prefill is compute-bound and decode is memory-bound holds for long prompts and batched prefills. It does not hold for a lone 64-token question. The property the cache uses is sequentiality: prefill pays the weight-stream once, and decode pays it T times.

Time lower bounds. A memory-bound kernel cannot beat B_w/BW . A compute-bound kernel cannot beat F/peak . Taking the max is the standard bound (Pope et al., 2023). For Llama 3.1 70B, batch one, BF16, on one H100,

$$\begin{aligned} t_{\text{mem}} &= 141 \text{ GB} / 3.35 \text{ TB/s} = 42 \text{ ms}, \\ t_{\text{comp}}^{\text{prefill}}(S=128) &= 17.6 \text{ TFLOP} / 989 \text{ TFLOP/s} = 18 \text{ ms}, \\ t_{\text{comp}}^{\text{dec}} &= 0.14 \text{ TFLOP} / 989 \text{ TFLOP/s} = 0.14 \text{ ms}. \end{aligned}$$

So a 128-token prefill and a single decode step have the *same* 42 ms floor. A 256-token completion has a 10.8 s floor. FP8 halves the weight footprint to 71 GB and halves those floors to 21 ms and 5.4 s; the ratio $T:1$ does not change. Tensor-parallel shards of the same model change the per-chip bytes and add all-reduce, but they do not change the fact that decode repeats the stream.

Table 4: Roofline lower bounds for one request, Llama 3.1 70B, $S = 128$, $T = 256$. “Chip-ms” at batch B is wall time of the batched kernel divided by B : the accelerator-time charged to the request. Prefill at $B = 32$ is compute-bound on H100 FP8 ($BS = 4096 > I^*$); decode is not.

	H100, batch 1		H100 FP8, batch 32	
	BF16	FP8	per batch	per request
Prefill, $S=128$	42 ms	21 ms	285 ms	8.9 ms
Decode, $T=256$	10.8 s	5.4 s	5.4 s	168 ms
Decode share of chip-time	99.6%	99.6%	95%	95%
CPU $e(Q)$ + ANN (typical)	1–10 ms, not on this chip			

TPU v5p yields the same qualitative bound with different constants. One chip has 96 GB of HBM, so a BF16 70B must be sharded across at least two chips; an FP8 70B fits on one. Per-chip bandwidth is 2.8 TB/s, slightly below H100, and I^* is lower (164), so short prefills are if anything more memory-bound. TPU v5e (16 GB, 0.82 TB/s) is a prefill-friendly small chip and would never hold this model alone. TPU v6e raises peak FLOPs faster than bandwidth ($I^* = 575$), which makes short prefills even more firmly a weight-stream. On every one of these parts, decode is the term a cache can delete.

A 700 W H100 burning 42 ms for prefill spends about 30 J. The same chip burning 5.4 s for an FP8 256-token decode spends about 3.8 kJ. Energy follows chip-time.

Retrieval embedder cost. MiniLM-L6 has $P_e \approx 2.3 \times 10^7$ parameters (Wang et al., 2020). Its forward cost $2P_e S \approx 5.8 \times 10^9$ FLOPs at $S = 128$ is $3000\times$ smaller than the 70B prefill. A 335M embedder is still $\sim 200\times$ smaller. Either one plus an HNSW probe of 10^7 – 10^9 vectors is a few milliseconds on the host (Malkov & Yashunin, 2020). Returning a 256-token cached string is a few kilobytes; inside a datacenter that is sub-millisecond. These terms matter for latency tails. They do not appear in the accelerator bill.

2.4 Implications for completion caching

Table 4 says that at chat lengths, decode is 95–99.6% of serving-model accelerator time. A hit that prefills Q , inspects $h(Q)$, and returns C^* therefore discards almost all of the request’s chip-time. A more precise gate that still skips decode is cheap: the extra prefill costs on the order of one decode token. The host-side embedder $e(Q)$ is cheaper still, and is the only way to leave the serving accelerators untouched. That last property matters when the fleet is prefill-saturated, or when prefill and decode are disaggregated onto different machines (Zhong et al., 2024). In a disaggregated stack a prefill-then-accept hit still occupies a prefill chip and still frees a decode chip. Decode chips scale with T and are hard to keep at high utilization; the cache’s value is the time it frees on those chips.

Those ratios give the three tiers of Section 3 a cost:

1. **Retrieval tier.** Compute $e(Q)$ on the host. If a cached C_i passes a conservative gate on $(e(Q), e(Q_i))$, return C_i . Serving-accelerator time: zero.
2. **Prefill tier.** Otherwise prefill Q , obtain $h(Q)$, and gate on the serving model’s own state. On accept, discard the KV cache and return C_i . Accelerator time: one weight-stream.
3. **Decode tier.** On reject, continue from the KV cache just written and generate $C(Q)$, optionally drafting from C_i . Accelerator time: the usual T streams. Insert $(Q, C(Q))$ if the request was cache-eligible.

Long-context and unique-document requests stay in the decode tier, and often should not be inserted: their cost is prefill, their Q will not recur, and $h(Q)$ represents a document the next user does not have. Context-bound intents (location, account, live search) are not portable completions; Section 3.5 reuses a tool plan rather than another user’s C_i . The napkin math is for portable decode-skips and for bound plans that avoid decode (short Q , longish C , repeated intents), where T weight-streams are the cost a hit avoids.

3 Completion caching

Section 2 priced the three places a request can leave the serving path: after a host-side retrieval, after a serving-model prefill, or after a full decode. This section specifies the policy that chooses among them. The objective is the number of decode-skips, subject to a bound on serving an unacceptable answer. An answer is unacceptable if it addresses the wrong question, or if it is correct only under another user’s context.

3.1 Cache records and the serve path

A completion cache is a set of records

$$\mathcal{D} = \{ r_i = (Q_i, C_i, e_i, h_i, [Q_i], \kappa_i, \pi_i, \mathcal{T}_i, m_i) \}.$$

$C_i = C(Q_i)$ at insertion; $e_i = e(Q_i)$ is the retrieval embedding; $h_i = h(Q_i)$ is the serving model’s last-token prefill state when it was stored (optional). $[Q_i]$ is a query equivalence-class representative (Section 3.2). κ_i is a binding class: **portable**, **bound**, or **ineligible**. For bound records, π_i is a tool plan (a program with holes) and $\mathcal{T}_i$ is a completion template with the same holes. Metadata m_i holds insertion time, model version, TTL, and an optional quality score.

Write $\mathcal{A}(Q, C)$ for the (unobserved) predicate that C is an acceptable reply to Q . Acceptability depends on intent and on context: the stored answer must be one this user, at this time and place, could have been given. The serve path is Algorithm 1; Figure 2 shows the control flow.

An exact string match is a candidate only after $n(\cdot)$ has folded trivial surface variation and BindClass has returned **portable**. Retrieval still returns a shortlist of $k > 1$ records. On portable traffic the gate is ternary: ACCEPT is calibrated so that served completions lie in $\mathcal{A}$; REJECT falls through to the decode that would have run anyway; DEFER spends a prefill, which Section 2 priced at about one decode token.

3.2 Equivalence classes and context binding

Equivalence. Write $n(Q)$ for a representative of the query’s equivalence class $[Q]$. The map n is a stack, cheapest first:

1. Unicode NFKC, case folding, strip trailing punctuation, collapse whitespace.
2. A small, closed rewrite list: contractions (*what’s* $\rightarrow$ *what is*), a handful of locative paraphrases (*near me* / *around me* / *nearby* / *close by*).
3. Optional learned equivalence: a classifier or a very conservative embedding threshold trained on labeled paraphrase pairs, used only to merge $[Q]$, never to authorize serving C_i .

Thus “What is the best Italian restaurant near me”, the same sentence with a question mark, “what’s the best Italian restaurant near me”, and “what are the best Italian restaurants around me” can share a class. The shared class is an index key. Serving C_i still requires BindClass and the gate, because the restaurant class is a single intent whose answer depends on who asked.

Algorithm 1 Serve a query against a completion cache.

Require: query Q , user/session context u , cache $\mathcal{D}$, embedder e , serving model, integers k, r

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1:  $[Q] \leftarrow n(Q)$  ▷ equivalence-class representative
2:  $\kappa \leftarrow \text{BindClass}(Q, u)$  ▷ portable | bound | ineligible; cheap, possibly learned
3: if  $\kappa = \text{ineligible}$  then
4:   return  $\text{Generate}(Q)$  ▷ do not insert
5: end if
6: if  $\kappa = \text{bound}$  then
7:   return  $\text{FillBound}(Q, u, [Q], \mathcal{D})$  ▷ tool plan + template; Section 3.5
8: end if
9: if  $\exists r \in \mathcal{D}$  with  $[Q_r] = [Q]$ , fresh, same model,  $\kappa_r = \text{portable}$  then
10:  if  $\text{CheapGate}(Q, u, r) = \text{ACCEPT}$  then
11:    return  $C_r$  ▷ gated even on exact text
12:  end if
13: end if
14:  $e_Q \leftarrow e(Q)$ ;  $\mathcal{N} \leftarrow \text{kNN}(e_Q, \mathcal{D}; k)$ 
15:  $(C^*, d) \leftarrow \text{PoolAndGate}(e_Q, \mathcal{N}, u; \text{space } e)$ 
16: if  $d = \text{ACCEPT}$  then
17:   return  $C^*$  ▷ retrieval tier
18: end if
19: if  $d = \text{REJECT}$  then
20:   return  $\text{MaybeInsert}(Q, \text{Generate}(Q), u)$ 
21: end if
22:  $(h_Q, \text{KV}) \leftarrow \text{Prefill}(Q)$  ▷  $d = \text{DEFER}$ 
23:  $(C^*, d) \leftarrow \text{PoolAndGate}(h_Q, \mathcal{N}, u; \text{space } h)$ 
24: if  $d = \text{ACCEPT}$  then
25:   discard KV; return  $C^*$ 
26: end if
27: return  $\text{MaybeInsert}(Q, \text{Decode}(\text{KV}; \text{draft } C^*), u)$ 

```

Binding. $\text{BindClass}(Q, u)$ assigns $\kappa \in \{\text{portable}, \text{bound}, \text{ineligible}\}$. The assignment is computed from the request’s tool list and from whether the provider’s per-user agent memory is read.

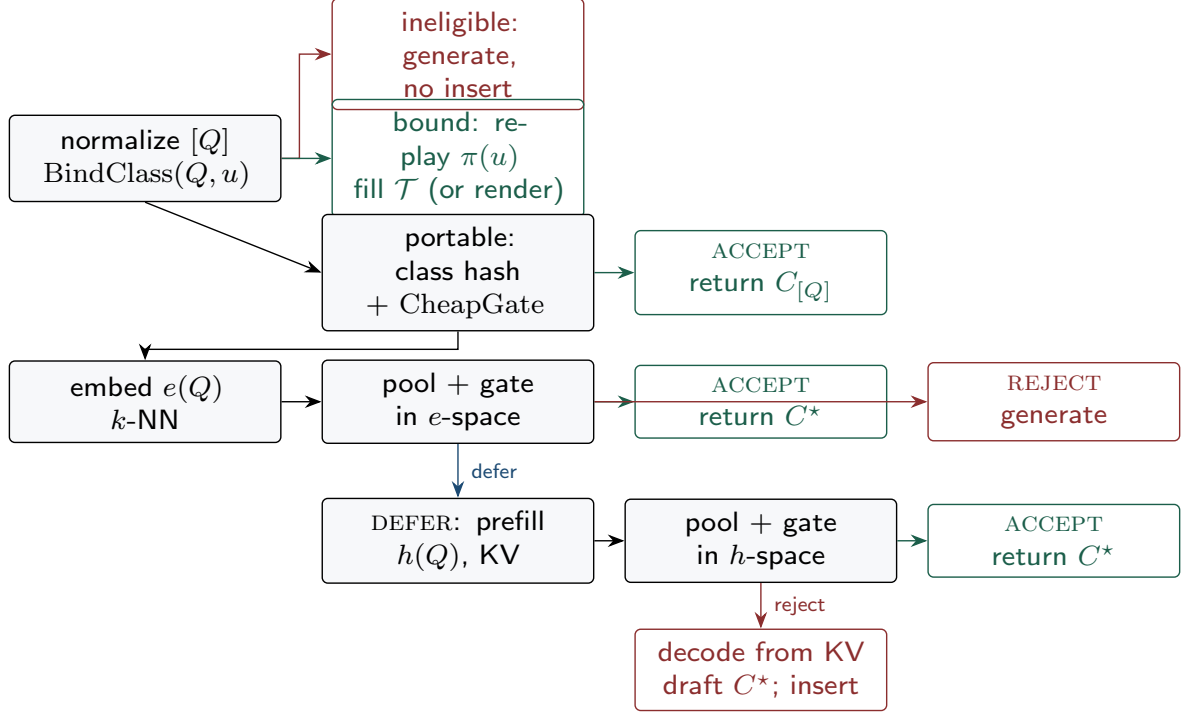
If the request does not invoke tools and does not read per-user agent memory, the completion does not depend on who asked. BindClass returns **portable**, and a stored C_i may be served. How-to questions, definitions, and other closed queries with no tool use fall here.

If the request does invoke tools, a second check inspects the tool names and, when available, their arguments for user-specific or otherwise non-universal context: location, weather, calendar, account state, the open document, a live local index. Presence of such a dependency yields **bound**: the tool plan and the wrap may be reused, the filled answer may not. Tools that do not introduce that dependency (a calculator, a public documentation lookup) do not by themselves force **bound**.

A read of per-user agent memory is treated as a non-universal context source even when no explicit tool call appears in the trace. A completion produced under another user’s memory is not an acceptable reply to Q . If the rest of the request has a reusable tool plan, the record can still be **bound**; otherwise it is **ineligible**.

Ineligible intents have neither a reusable answer nor a reusable plan: a unique pasted file with no tool schema, a jailbreak, or a request the provider has chosen not to cache.

At insertion, κ_i is taken from the generation trace (tools invoked, memory read, whether any tool



Green leaves skip serving-model decode. Bound fill-in may still call tools. Exact class-hash still goes through CheapGate.

Figure 2: Serve path. BindClass runs before any completion is reused, from the tool list and from whether per-user agent memory is read. Portable queries may return a cached C^* after CheapGate, including on an equivalence-class hash. Bound queries replay a tool plan into a template under this user’s context. Ineligible queries decode and are not stored. DEFER spends a serving-model prefill to decide again in h -space.

result was non-universal). At serve time, BindClass(Q, u) predicts that label from Q , u , and any tool schema already stored for $[Q]$. The predictor can start as a lexicon and regular expressions on Q and on the tool list (“near me”, “my”, “I have”, “today”, “this file”, session identifiers) and can later become a small learned model. Its cost is in the same budget as $e(Q)$: host-side, milliseconds. A false **portable** assignment serves one user another user’s restaurant, medical detail, or file, so the predictor is conservative toward **bound** and **ineligible**. A false **bound** still skips decode if a tool plan exists.

A stored record carries its own κ_i . A portable completion is reusable only for a portable query in the same $[Q]$ (or a pooling accept, Section 3.4). A bound record exports π_i and $\mathcal{T}_i$, not C_i .

3.3 Gating

A false accept serves the wrong answer. A false reject only costs a decode. Gates therefore constrain $\Pr(\neg \mathcal{A} \mid \text{ACCEPT})$; hit rate is whatever remains after that constraint. CheapGate is the hash-path filter: binding class, freshness, model version, and a check against u . Matching tokens are not sufficient to accept.

Let x_Q be the query representation in the current space (e_Q at the retrieval tier, h_Q at the prefill tier) and let $\mathcal{N} = \{r_i\}_{i=1}^k$ be the shortlist. The gate g reads x_Q , the matching fields of $\mathcal{N}$, and

u , and returns $(\hat{C}, d)$ with $d \in \{\text{ACCEPT}, \text{DEFER}, \text{REJECT}\}$. Raw-embedding cosine scores topical relatedness, which is the France/Spain failure mode from Section 1.1. The features used by g are:

- **Binding compatibility.** If κ or κ_i is not **portable**, g cannot accept C_i . Bound shortlists are handed to Section 3.5.
- **Projected residual ρ** and projected neighbor i^* from top- k projective pooling (Section 3.4). A small residual means Q lies in the local paraphrase span of $\mathcal{N}$. A large residual means the shortlist is a topical near-miss.
- **Completion consensus γ :** do the k stored answers agree with each other? A mixed shortlist is a collision of intents that happen to sit nearby in representation space.
- **Freshness and version.** A record with an expired TTL, or a model version other than the one now serving, cannot be accepted as C_i . It may still be used as a draft or a template.

A simple, deployable g on portable traffic is a threshold rule on (ρ, γ) after those hard filters:

$$d = \begin{cases} \text{ACCEPT} & \text{if } \rho \leq \tau_\rho^{\text{hi}}, \gamma \geq \tau_\gamma, \kappa = \kappa_i = \text{portable}, \\ \text{REJECT} & \text{if } \rho > \tau_\rho^{\text{lo}} \text{ or a hard filter fails,} \\ \text{DEFER} & \text{otherwise, and only at the retrieval tier.} \end{cases} \quad (7)$$

At the prefill tier there is no further deferral: DEFER collapses to REJECT and decode starts from the KV cache just written. The two thresholds $\tau_\rho^{\text{hi}} < \tau_\rho^{\text{lo}}$ leave a band in which e -space is unsure and a prefill in h -space is justified. Both τ and τ_γ are calibrated offline on labeled (paraphrase, near-miss, bound) triples so that the empirical false-accept rate stays under a chosen ε (for example 10^{-3}). A learned classifier on the same features can replace (7); the calibration target does not change.

The served completion on ACCEPT is the output of the pooling rule below, not necessarily the nearest neighbor.

3.4 Top- k projective pooling

Nearest-neighbor caches treat the shortlist as a ranked list and take the head. This ignores two properties of portable intents. Many such intents have several acceptable answers, so a unique neighbor leaves hits on the table. Paraphrase variation is approximately low-rank in a local neighborhood, while the coordinates that change the answer (a version number, an entity, a negation) are often the ones that stick out of that neighborhood. Top- k projective pooling uses the shortlist as a local basis rather than as a leaderboard. It is a rule for portable records. Bound records are pooled in plan-and-template space, not in C_i space (Section 3.5).

Local paraphrase subspace. Let $x_1, \dots, x_k$ be the shortlist representations in the current space and $\bar{x} = \frac{1}{k} \sum_i x_i$. Form the centered matrix $X = [x_1 - \bar{x} \mid \dots \mid x_k - \bar{x}]$ and take the top r left singular vectors U_r (typically $r = \min(3, k - 1)$). Write $P = U_r U_r^\top$ for the projector onto that span. The query residual is

$$\rho(Q, \mathcal{N}) = \frac{\|(I - P)(x_Q - \bar{x})\|}{\|x_Q - \bar{x}\| + \epsilon}. \quad (8)$$

A small ρ means Q is a displacement inside the neighborhood, the usual signature of paraphrase. A large ρ means Q has a component the neighbors do not share. That is the signature of a near-miss

that cosine still ranks highly (“Python 3.11” versus “3.12” when both sit near a pile of “install Python” questions).

The projected neighbor is

$$i^* = \arg \min_{i \leq k} \|P(x_Q - \bar{x}) - P(x_i - \bar{x})\|. \quad (9)$$

Distances in the span compare phrasing; the residual in (8) is what decides whether that comparison is about the same question.

Consensus. Let $\phi(C_i)$ be a cheap embedding of the stored completions (the same e , or even a bag-of-tokens hash). Cluster $\{\phi(C_i)\}$ with a single-linkage cutoff. Write γ for the fraction of the shortlist that lands in the largest cluster, and write $\mathcal{N}_{\text{maj}}$ for that cluster. If γ is small, the neighborhood is mixed: the local span includes more than one answer, and accepting C_{i^*} would return an arbitrary member of that set. Those shortlists never accept at the retrieval tier. They may still accept at the prefill tier, where $h(Q)$ can break the tie, or they fall through to decode.

The pooled completion. On ACCEPT,

$$C^* = C_{j^*}, \quad j^* = \arg \min_{r_j \in \mathcal{N}_{\text{maj}}} \|P(x_Q - \bar{x}) - P(x_j - \bar{x})\|, \quad (10)$$

breaking remaining ties by freshness and stored quality. We return one stored completion from the majority cluster. Blending portable completions would emit text that no logged generation produced. The shortlist is used to decide whether an acceptable stored answer is present and which one to return.

The constants k and r are set by two failure modes. If k is too small, the estimated span U_r is noisy and ρ is meaningless. If k is too large, the shortlist reaches into neighboring intents, γ drops, and the gate defers or rejects. If r is too large, P becomes a near-identity and ρ cannot see a near-miss. The useful regime is a small neighborhood that is still large enough to estimate a two- or three-dimensional paraphrase plane: $k \in [8, 32]$, $r \in \{2, 3\}$ is the intended operating point. Those constants are for the designer to sweep; they are not claims about a deployed system.

3.5 Bound queries: tool plans, templates, and non-AR fill-in

A bound intent has a reusable program and a reusable presentation shape. Caching C_i copies another user’s result into this user’s thread. The cached object is the program, not the filled answer.

Tool plans. When a bound request is generated the first time, the serving model typically emits one or more tool calls (maps search, catalog lookup, calendar, code execution) and then a natural-language wrap. MaybeInsert stores, in addition to or instead of C_i , a plan π : a sequence of tool names and argument *templates* whose holes are bound at serve time from u (user coordinates, account id, “now”, the active document). Replaying $\pi(u)$ is ordinary RPC, not decode. The restaurant class becomes `places.search(cuisine=italian, near = u.geo)`, not “try Mario’s on Oak Street.”

Templates. The wrap around a tool result is low-entropy given the slots. From a shortlist of bound records in $[Q]$ we extract a completion template $\mathcal{T}$ with holes aligned to π ’s outputs: *The*

highest-rated Italian place near you is {name} ({rating}) at {addr}. Extraction can be heuristic (longest common subsequence with slots masked) or learned. FillBound then does

$$C = \mathcal{T}(\pi(u)), \quad (11)$$

string interpolation, not generation. If several templates in the class disagree, take the majority skeleton or fall through to decode. Consensus here is consensus of *shapes*, not of filled-in names.

Non-autoregressive fill-in. Interpolation is brittle when the wrap is not quite a template: a slightly different tone, a hedge, an extra clause. A small conditional generator that maps $(\mathcal{T}, \pi(u))$ to a surface string, trained offline on logged (template, slots, completion) triples, can sit in that gap. Discrete or latent diffusion models for text are one candidate: they are not token-serial in the serving-model sense, and they can be run on a cheap device next to the router. The renderer is optional. It sees u only through the already-fetched slots, and it is not permitted to invent a restaurant. If a confidence check on the renderer fails, the request falls through to serving-model decode.

FillBound is a fourth exit from the serving path, cheaper than prefill when the plan hits. The only reused tokens are those that do not mention a person, and the only inserted facts were fetched for this request.

3.6 Decode, draft, and insert

On a portable miss, the serving model generates $C(Q)$ as usual. Two decisions remain: whether the retrieved completion can cheapen that generate, and what to write back to $\mathcal{D}$.

Speculative draft. If PoolAndGate produced a C^* that it would not accept as an answer (a high cosine with a large residual, or a mixed shortlist), C^* is still a plausible draft. Retrieval-based speculative decoding (He et al., 2024; Leviathan et al., 2023) verifies a draft in one forward pass per chunk and keeps a prefix that matches the model’s own distribution. Using a cached completion as that draft costs no extra model and cannot change the output. A rejected hit therefore becomes a partial decode-skip rather than a full miss.

Insertion. MaybeInsert is binding-class-specific. Portable, short-to-medium, non-document Q store $(Q, C, e, h, [Q], \text{portable}, m)$. Bound Q store $[Q]$, $\kappa = \text{bound}$, the tool plan π , and the template $\mathcal{T}$; they do not store a filled C as a reusable answer. Ineligible Q store nothing. Time-sensitive portable facts (sports scores, prices) get a TTL or they are reclassified as bound and given a refresh tool. Model version is part of the key for h_i and C_i ; e_i , $[Q]$, π , and $\mathcal{T}$ can be rebuilt or kept across a bump. Eviction is ordinary: LRU or LFU on the retrieval index, with a floor for records that have already paid for themselves in decode-skips.

Storing other users’ locations and unique documents raises occupancy and makes hits unsafe. The cost model in Section 2 applies to portable decode-skips and to bound plans that avoid decode. Those are the records MaybeInsert stores.

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